OJAS PAL

ojaspal.dev@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI & Data Science undergraduate specializing in building RAG pipelines and deploying LLM applications on Google Cloud. Experienced across the full data-to-deployment cycle: data preprocessing, vector search, model fine-tuning, and serving. Focused on shipping reliable, production-style GenAI systems.

EDUCATION

University School of Automation and Robotics (USAR), GGSIPU East Campus, Delhi **2024 – 2028**
B.Tech in Artificial Intelligence & Data Science (AI-DS)

TECHNICAL SKILLS

Languages: Python, C++

ML & Data: scikit-learn, Pandas, NumPy, NLTK

Deep Learning & Vision: PyTorch, TensorFlow, Keras, OpenCV

Concepts: Computer Vision, NLP, Recommender Systems, Feature Engineering

AI & GenAI Frameworks: LangChain, Groq API, Gemini API, RAG (Retrieval-Augmented Generation)

Databases & Cloud: Pinecone Vector Database, Firebase Authentication, Cloud Firestore, Google Cloud Run, Google Cloud Secret Manager

Backend & UI: Flask, REST APIs, Streamlit

Developer Tools: Git, GitHub, Jupyter Notebook, Google Cloud IAM

PROJECTS

ML-DL Portfolio: 11 Models in One Streamlit App | GitHub **October 2026**
Python, Streamlit, Scikit-Learn, CatBoost, PyTorch, TensorFlow, Keras, OpenCV

- Rebuilt 11 old Jupyter notebooks (8 ML, 3 DL) as one multi-page Streamlit app. Models are saved once as joblib, .pth and .keras files and loaded with @st.cache_resource, so pages don't retrain on load.
- Replaced an ~8 GB precomputed similarity matrix (32,833 songs) with per-query cosine similarity, so the Spotify page fits Streamlit Community Cloud's 2.7 GB memory limit.
- Fine-tuned EfficientNet-B0 in two stages on 46 breeds merged from Oxford-IIIT Pet and Stanford Dogs. It reached 93.6% validation accuracy and predicts in 25–30 ms on Streamlit Cloud's CPU.

Medical RAG Chatbot | GitHub **September 2026**
Python, LangChain, Groq API, Pinecone, HuggingFace, Flask, Google Cloud Run

- Split a 637-page medical encyclopedia into 5,800+ chunks, embedded them with bge-small-en-v1.5 (384-dim) and indexed them in Pinecone. Top-2 lookups return in under 45 ms.
- Passed the retrieved passages to gpt-oss-20b on Groq through LangChain, with a strict prompt that keeps answers tied to the source text and blocks non-medical questions.
- Deployed on Cloud Run (2 GiB, 2 vCPU, 8 threads). Lazy-loading the embedding model gives sub-second server startup, so health checks pass.

Personal Gemini Journal | GitHub **September 2026**
Gemini API, Firestore, Firebase Auth, Google Cloud Run, Secret Manager

- Wrote Firebase Auth middleware that verifies the ID token, with a revocation check, on all 9 REST endpoints. Entries live under /users/{uid}, so one user can't read another's data.
- Fetched the Gemini key from Secret Manager at runtime with a 1-hour in-memory cache. The Cloud Run service account has only 3 IAM roles and no keys sit in the repo.
- Built "Ask My Journal", a search over past entries using Gemini embeddings plus keyword scoring. Each reply also pulls the last 3 entries as context with zero extra API calls.

CERTIFICATIONS

- Google Cloud Gen AI Academy, APAC Edition, Cohort 3 (Hack2Skill)
- IBM SkillsBuild AI Automation & Intelligent Solutions Internship